IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

EEE/EIE PART III/IV: MEng, BEng and ACGI

DIGITAL SIGNAL PROCESSING

Thursday, 11 December 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : T. Stathaki
Second Marker(s) : W. Dai

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1. (a) Consider the causal real-coefficient transfer function

$$H_M(z) = \frac{\sum_{n=0}^{n=M} b[n]z^{-n}}{\sum_{n=0}^{n=M} a[n]z^{-n}}, M > 0$$

(i) Prove that when the coefficients of the numerator and denominator satisfy the symmetry condition $a[n] = b[M - n]$, the transfer function $H_M(z)$ corresponds to an allpass filter. **[3]**

(ii) Comment on the locations of the zeros and the poles of a causal real-coefficient stable allpass filter's transfer function within the z −plane, with respect to the unit circle. Justify your answer. **[3]**

(iii) Find the part of the phase response of an allpass transfer function due to its zeros, in terms of the part of the phase response due to its poles. **[3]**

(iv) Explain how multiplying $H_M(z)$ by z^{-N} affects its amplitude and phase responses and discuss what happens when the integer N becomes large. **[3]**

(b) Consider a second-order causal real-coefficient stable allpass transfer function $H(z)$.

(i) Provide the general expression for $H(z)$. **[3]**

(ii) Find the phase response associated with $H(z)$. **[3]**

(c) (i) Using a first-order non-minimum-phase IIR filter, demonstrate that it can be expressed as the product of a minimum-phase IIR filter and an allpass filter. **[4]**

(ii) Compare the average group delay of a first-order non-minimum-phase IIR filter with that of its minimum-phase counterpart. **[3]**

2. (a) (i) Show that a **Type III** linear-phase FIR transfer function $H(z)$ of length $2M+1$ can be expressed as follows:

$$H(z) = z^{-M}\left(\sum_{k=0}^{M-1} h[k](z^{M-k} - z^{k-M})\right) \quad (1)$$

[4]

(ii) Determine the phase of the expression enclosed in the brackets of equation (1) over the interval $0 < \omega < \pi$. **[4]**

(b) By using the relation

$$z^r - z^{-r} = (z - z^{-1})U_{r-1}\left(\frac{z+z^{-1}}{2}\right) \Rightarrow$$

$$U_{r-1}\left(\frac{z+z^{-1}}{2}\right) = \frac{z^r - z^{-r}}{z - z^{-1}}$$

where $U_r(x)$ is the r −th order **Chebyshev polynomial of the second kind** in x, express $H(z)$ of (a)(i) above in the form

$$H(z) = z^{-M}(z - z^{-1})\sum_{n=0}^{M-1} a[n]\left(\frac{z+z^{-1}}{2}\right)^n$$

in the case where $M = 4$. Determine the coefficients of $a[n]$ as functions $h[n]$ for the case $M = 4$.

The Chebyshev polynomials of the second kind in x satisfy the following recursive relationship:

$U_r(x) = 2xU_{r-1}(x) - U_{r-2}(x), r \geq 2$

$U_0(x) = 1, U_1(x) = 2x$

[8]

(c) (i) Show that a linear-phase FIR filter with symmetric impulse response and length M satisfies the Mirror-Image Polynomial (MIP) relationship given below. **[3]**

$$H(z) = z^{-(M-1)}H(z^{-1})$$

(ii) Show that a linear-phase FIR filter with antisymmetric impulse response and length M satisfies the Anti–Mirror-Image Polynomial (AIP) relationship given below. **[3]**

$$H(z) = -z^{-(M-1)}H(z^{-1})$$

(iii) Using the Mirror-Image Polynomial (MIP) and Anti–Mirror-Image Polynomial (AIP) relationships, show that bandstop filters can only be realised with Type I linear-phase FIR filters. **[3]**

3. (a) (i) Explain using mathematical relations, where each subset of the z −plane is mapped on the s −plane through the bilinear transformation. **[5]**

(ii) Prove that the relationship between the continuous-time angular frequency and the discrete-time angular frequency is non-linear. **[5]**

(b) A real-coefficient lowpass digital IIR filter has transfer function $H_L(z)$ and cutoff frequency ω_c. A new filter is obtained by applying the transformation $z \mapsto F(z) = \frac{az+b}{cz+d}$, giving the transformed transfer function $H_H(z) = H_L(F(z))$ which is required to be a real-coefficient highpass filter of the same order as $H_L(z)$, but with a different cutoff frequency. Using the constraints $H_H(e^{j\pi}) = H_L(e^{j0}) = 1$ and $H_H(e^{j0}) = H_L(e^{j\pi}) = 0$, derive the relationships among the parameters a, b, c, d and state what type of system the mapping $F(z)$ represents. **[5]**

(c) (i) Consider the Finite Impulse Response (FIR) transfer functions $G_1(z) = \frac{1}{4}(z + 2 + z^{-1})$ and $G_2(z) = \frac{1}{4}(-z + 2 - z^{-1})$. Explain what type of filters $G_1(z)$ and $G_2(z)$ represent and justify your answer. **[5]**

(ii) Find the average group delay of $G_1(z)$ and $G_2(z)$. **[5]**

4. (a) Consider the multirate structure of **Figure 1** below, where $H_0(z), H_1(z)$ and $H_2(z)$ are ideal zero phase real-coefficient lowpass, bandpass and highpass filters respectively, with frequency responses as follows:

$$H_0(e^{j\omega}) = u(\omega) - u(\omega - \frac{\pi}{3})$$

$$H_1(e^{j\omega}) = u(\omega - \frac{\pi}{3}) - u(\omega - \frac{2\pi}{3})$$

$$H_2(e^{j\omega}) = u(\omega - \frac{2\pi}{3}) - u(\omega - \pi)$$

The function $u(\omega)$ is the well-known unit step function, defined as

$$u(\omega) = \begin{cases} 1 & \omega \geq 0 \\ 0 & \text{elsewhere} \end{cases}$$

If the input is a real sequence with a Discrete Time Fourier Transform

$$X(e^{j\omega}) = (-\frac{9\omega^2}{\pi^2} + 1)\left(u(\omega) - u\left(\omega - \frac{\pi}{3}\right)\right)$$

sketch the Discrete Time Fourier Transform of the outputs $y_0[n], y_1[n]$ and $y_2[n]$.
Notice that all frequency domain representations above are provided only within the range $[0, \pi]$. **[15]**

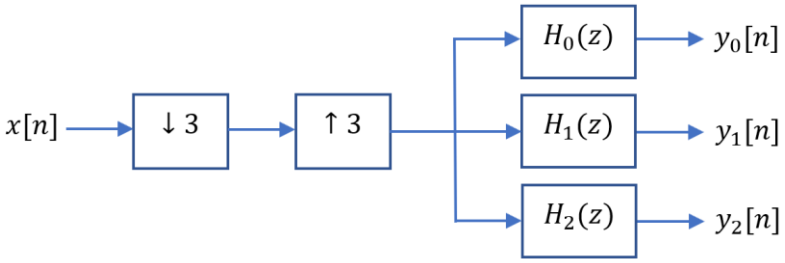


Figure 1

(b) Develop an expression for the output $y[n]$ as a function of the input $x[n]$ for the multirate structure of **Figure 2** below. **[10]**

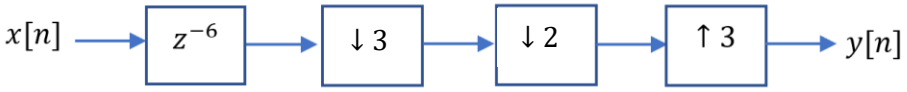


Figure 2